DESY MAHARANI

AI/ML ENGINEER

LLM Applications | AWS Cloud | Full-Stack (React Native, TypeScript)

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PROFESSIONAL SUMMARY

AI Engineer with 1.5+ years of experience building LLM-powered systems and shipping them to production on AWS. Most recently at EnsoNex, a creator-powered user acquisition platform serving major gaming publishers across the USA, Europe, Middle East, Japan, and Korea. Joined as a founding member of the Malaysia engineering team to architect and automate workflows previously handled entirely manually — from creator discovery and outreach to campaign management and payment processing. Designed and built, with full ownership, a serverless LLM pipeline that tags 16,000+ creator profiles into 300+ games and 200+ genres with confidence scoring, reducing tagging time from ~3 minutes to under 1 second per creator. Previously contributed to LLM-powered essay marking and exam generation tools at Pandai, using DSPy for prompt optimization and Langfuse for evaluation and monitoring. Full-stack background in React Native, TypeScript, and GraphQL, so AI features ship inside real products, including a multi-portal SaaS platform serving 40+ brands.

TECHNICAL SKILLS

AI & ML: LLM APIs, AWS Bedrock, DSPy, Langfuse, RAG, TensorFlow, PyTorch

Programming Languages: Python, TypeScript, JavaScript, SQL, Java

Cloud & Infrastructure: AWS Amplify Gen 2, Lambda, S3, Cognito, DynamoDB, Bedrock, IAM, CloudWatch

Backend & APIs: GraphQL (AWS AppSync), REST APIs, Node.js, FastAPI

Frontend & Mobile: React Native, React, Expo Router, Tailwind CSS, NativeWind, Xcode, Android Studio

Databases: AWS DynamoDB (NoSQL), PostgreSQL, MySQL

Tools & Practices: Git, GitHub, VS Code, Postman, Figma, Agile/Scrum, Cross-platform Development

PROFESSIONAL EXPERIENCE

AI Full-Stack Software Engineer | EnsoNex (DMSA Global) Sdn. Bhd.

Jan - Sept 2026

Cyberjaya, Malaysia • Creator-powered user acquisition platform serving major gaming publishers, connecting brands, agencies, and creators through AI-assisted campaign management and creator discovery.

- Designed and built an **AI-Powered Creator Tagging System** on AWS Bedrock, Lambda, GraphQL, and DynamoDB with the YouTube Data API — tagging **16,000+ creator profiles** sourced by the platform's in-house discovery system into **300+ games, 200+ genres, and 30+ content styles** with Confidence Score logic, reducing manual tagging from ~3 minutes to **under 1 second per creator (80 per batch).**
- Built the **Creator and Marketer/Brand portals** full-stack on a greenfield SaaS platform of **4 interconnected portals** serving **40+ brands,** including authentication, dashboards, campaign creation and management, creator discovery, onboarding verification, and analytics — using AWS Amplify Gen 2, AppSync, Lambda, Cognito, S3, and DynamoDB, and contributed features to the internal Staff portal.
- Built the **end-to-end campaign lifecycle** (invitation, multi-round negotiation, deliverables, document signing, submission) powered by GraphQL APIs and DynamoDB, supporting **30+ active campaigns** and digitizing workflows previously managed manually across **700+ campaigns.**
- Delivered **cross-platform UI across web, iOS, and Android** (React Native) with **dark/light theming** and responsive layouts for phones, tablets (iPad), and foldables, and contributed to a shared library of **~76 reusable components** used across all 4 portals to keep features consistent and speed up development.
- Wrote and tested **30+ FAQ documents** (SOPs, FAQs, policies) powering an internal **AI knowledge chat assistant** (AWS Bedrock, Amazon Nova, S3, Lambda) for the Staff and Marketer portals, handling **150+ queries per week** and **reducing support tickets by ~55%.**
- Collaborated cross-functionally with product managers, designers, and engineers to translate evolving business requirements into scalable technical solutions, and produced **system architecture diagrams and onboarding tutorials** for the platform.

AI Engineer Intern | Pandai Education Sdn. Bhd. Mar - Aug 2025
KL, Malaysia • AI-powered EdTech platform delivering personalized learning and automated assessment

- Developed LLM-powered applications, **integrating DSPy for prompt optimization** and **Langfuse for observability**.
- Contributed to AI-driven **automated essay marking and exam scheme** generation systems, improving assessment efficiency.
- Contributed to continuous **LLM evaluation and monitoring**, including **Langfuse auto-alerting**, to help improve accuracy and reduce false positives/negatives.
- Contributed to MagicAI Tools, Pandai App integrations, and the API Builder.
- Worked within an Agile/Scrum workflow, participating in sprint planning sessions, managing code through Git version control, and conducting peer code reviews across the engineering team.

AI Data Annotator | Lifewood Data Technology Sdn. Bhd. Aug - Dec 2025
Cyberjaya, Malaysia • Global AI data services company providing high-quality data annotation and multimodal datasets for autonomous driving, smart city, aviation, computer vision and machine learning applications.

- Supported autonomous driving, smart city, and aviation AI projects by processing and validating large-scale multimodal datasets (image, video, audio, LiDAR, sensor) with **99% data accuracy**.
- Built high-quality machine learning datasets by transforming unstructured data into structured, production-ready formats, accelerating AI model training, validation, and deployment.

PRODUCT & ENGINEERING HIGHLIGHTS

AI-Powered Creator Tagging System *AWS Lambda • GraphQL • LLM APIs • YouTube Data API • DynamoDB*
Designed and implemented an automated AI data enrichment pipeline that analyzes creator content and enriches creator profiles with structured game genres, creator styles, and game metadata.

Campaign Management & Negotiation Platform *React Native • GraphQL • TypeScript*
Built an end-to-end campaign workflow covering invitation, offer negotiation, deliverables, document signing, submission, settlement, and real-time workflow synchronization.

Smart-Bin (Final Year Project) *Deep Learning • AI • Computer Vision • IoT*
Developed an intelligent waste management system achieving 96% waste classification accuracy, recognized by Biji-Biji Initiative Malaysia and the U.S. Embassy in Malaysia.

EDUCATION

Bachelor of Computer Science (Data Science) 2022 - 2025
Albukhary International University, *Malaysia*

- Full Scholarship from Albukhary Foundation, Malaysia & Full Scholarship from Baznas Indonesia.
- Dean's List Awards.
- Gold Award, The Academy International Gifted Enrichment Research and Innovation 2024.
- 4th Best Paper Award at the 3rd International Conference on Information Technology and Security 2024.

PUBLICATIONS

Intelligent Waste Segregation System Using Convolutional Neural Networks for Deep Learning Applications
IC-ITECHS | DOI: 10.32664/ic-itechs.v5i1.1579 | December 2024

Performance Comparison of Machine Learning Models to Reduce Misdiagnosis Rates in Psychiatric Disorder Using EEG Dataset
Journal of Data Analytics, Information, and Computer Science | DOI: 10.59407/jdaics.v1i3.967 | July 2024

CERTIFICATIONS & TRAINING

AWS Skill Builder — Generative AI & Machine Learning Specialization
AWS Bedrock • AWS SageMaker AI • AWS Developing Generative Artificial Intelligence Solutions • AWS Developing Machine Learning Solutions • AWS Essentials of Prompt Engineering

AWS Academy — Machine Learning Foundations

AWARDS & HONORS

- **Awardee of Cendekia Baznas Indonesia Scholarship (2022-2025)**
- **Awardee of Albukhary Foundation Malaysia Scholarship (2022-2025)**
- **Awardee of Acceleration/Fast-Track Program in Senior High School (2018-2020)**
- **Awardee of Acceleration/Fast-Track Program in Junior High School (2016-2018)**